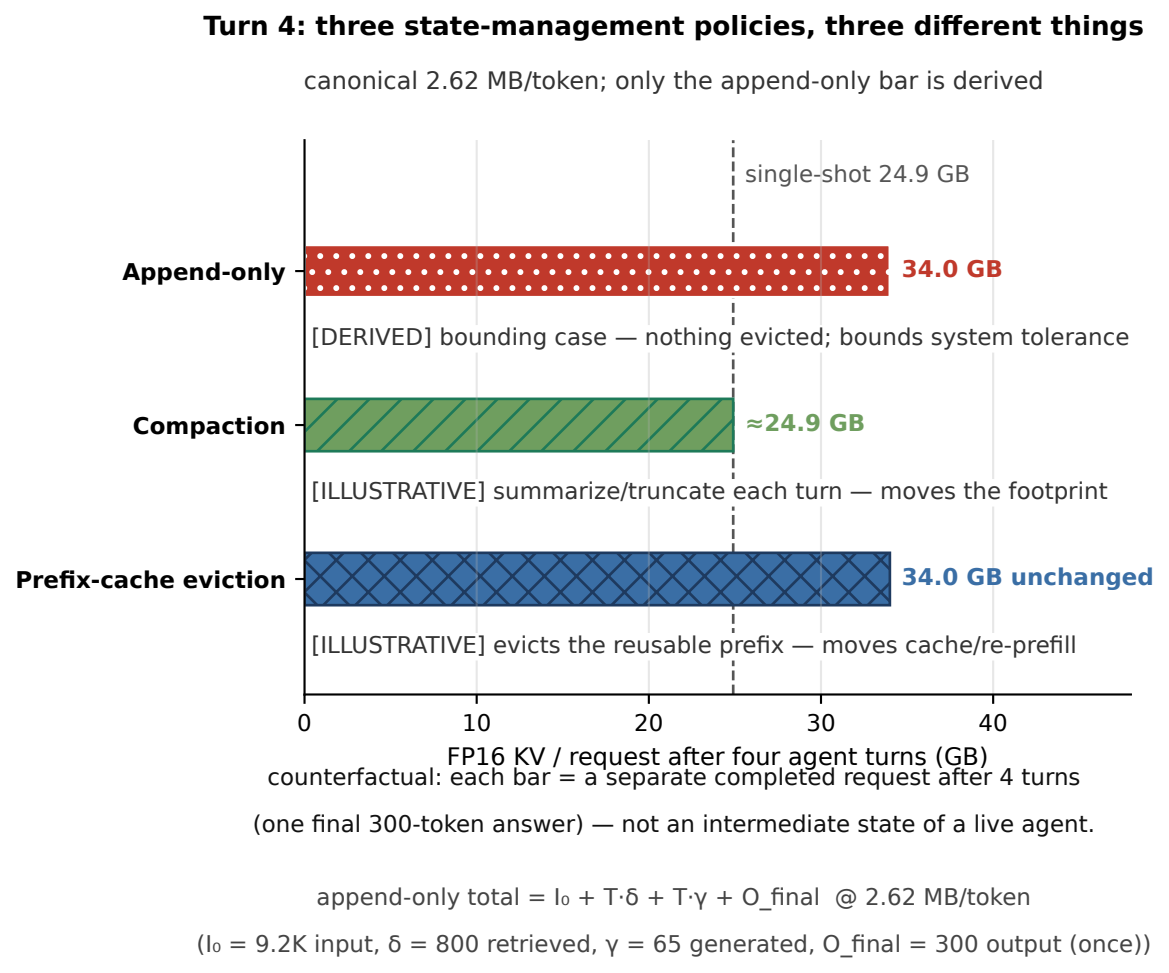
Turn 4: three state-management policies, three different things
canonical 2.62 MB/token; only the append-only bar is derived
single-shot 24.9 GB
Append-only
34.0 GB
[DERIVED] bounding case — nothing evicted; bounds system tolerance
Compaction
≈24.9 GB
[ILLUSTRATIVE] summarize/truncate each turn — moves the footprint
Prefix-cache eviction
34.0 GB unchanged
[ILLUSTRATIVE] evicts the reusable prefix — moves cache/re-prefill
0
10
20
30
40
FP16 KV / request after four agent turns (GB)
counterfactual: each bar = a separate completed request after 4 turns
(one final 300-token answer) — not an intermediate state of a live agent.
append-only total = I₀ + T·δ + T·γ + O_final @ 2.62 MB/token
(I₀ = 9.2K input, δ = 800 retrieved, γ = 65 generated, O_final = 300 output (once))